
ISOMERS
Constitutional
/ Structural
Same molecular formula, different connectivity
Types: Chain · Positional · Functional Group
e.g. n-butane vs isobutane (chain isomers)
Conformational
Conformers
Same molecule · bond rotation only
Interconvert freely · no bond breaking required
Newman projections: staggered · eclipsed · gauche · anti
Ring conformers: chair · boat · twist-boat
Stereo
Isomers
Same formula, same connectivity, different spatial arrangement
Require bond breaking (or reflection) to interconvert
Configurational
Isomers
Require bond breaking to change configuration; includes optical AND geometric isomers
Enantiomers
Non-superimposable mirror images
Opposite R/S at EVERY chiral center
Same physical properties (bp, mp, solubility, density)
Opposite optical rotation direction (+/−)
R / S
Absolute Config.
CIP Priority Rules
sp³ C with 4 different groups = chiral center
Rank substituents 1→4 (highest atomic # = 1)
R = clockwise (rectus)
S = counterclockwise (sinister)
D / L
Relative Config.
Fischer Convention
D = OH on right at reference carbon
L = OH on left at reference carbon
Most natural sugars: D-form
Most natural amino acids: L-form
(+) / (−)
Optical Rotation
Measured by polarimetry
(+) dextrorotatory = CW rotation of light
(−) levorotatory = CCW rotation
NOT correlated with R/S or D/L!
Meso
Compound
Has chiral centers, BUT achiral overall
Internal plane of symmetry
Optical rotation = 0 (inactive)
e.g. meso-tartaric acid
Racemic
Mixture
50:50 mix of enantiomers
Net optical rotation = 0
Cannot separate by standard physical methods
(need chiral column or resolving agent)
R/S · D/L · (+)/(−) are INDEPENDENT — each must be determined separately!
Max stereoisomers = 2ⁿ (n = # chiral centers); fewer if meso compound(s) exist
Diastereomers
NOT mirror images
Opposite R/S at ONLY SOME chiral centers
Different physical properties — can be separated!
Geometric
Isomers
Restricted rotation (C=C double bond or ring)
Cannot interconvert without breaking π bond
E/Z = preferred IUPAC; cis/trans = H-based (older)
E / Z
(CIP-based)
E (Entgegen) = higher priority groups on OPPOSITE sides of double bond
Z (Zusammen) = higher priority groups on the SAME side of double bond
Assign using CIP rules (same as R/S)
cis / trans
(H-based)
cis = same groups on same side
trans = same groups on opposite side
Only reliable for H-bearing carbons
Use E/Z when no H on double bond carbon
Epimers
Differ at ONLY ONE chiral center
(always diastereomers)
D-Glucose vs D-Galactose → C4 epimers
D-Glucose vs D-Mannose → C2 epimers
Anomers
Epimers at the anomeric carbon (C1)
Formed during ring closure of open-chain sugars
α = OH axial (below ring plane)
β = OH equatorial (above ring plane)
Interconvert in solution via mutarotation
LEGEND
▱ notation / designation ⬡ special case ◯ concept node ⬢ key exam rule ┈┈ conformers (not true isomers)
Purple = Configurational hierarchy Teal = Enantiomers section Red = Diastereomers section Orange = Constitutional Gray dashed = Conformational